

FLEET RULE OF THE ROAD TEST FOR JAN 2018

This test is based on: The International Regulations for Preventing Collisions at Sea (IRPCS) (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(G) Vol II, NP735 and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

PART 1

1. How should a vessel normally join or leave a TSS? **(R10b(iii))(1)**

2. Describe a masthead light (colour, arc of visibility and position on the ship). **(R21a)(1)**

3. Nothing in these Rules shall exonerate any vessel, owner, master or crew thereof from what? **(R2a)(1)**

4. The Rules regarding lookout, safe speed, risk of collision, action to avoid collision, narrow channels and Traffic Separation Schemes apply in what condition of visibility? **(R4)(1)**

5. Describe the lights displayed by a partly submerged vessel or object, or a combination of such vessels or objects being towed if less than 100m in length. **(R24g(i)&(ii))(2)**

6. Describe the lights, including any additional lights that may be displayed by a sailing vessel underway, if greater than 20m in length.(R25b&c)(2)

7. What does the term “vessel not under command” mean? (R3f)(1)

8. In construing and complying with these Rules, due regard shall be had to what?(R2b)(2)

9. Describe the sound signals, including time intervals, required for a vessel at anchor in restricted visibility, if greater than 100m. (R35g)(2)

10. What does the term prolonged blast mean?(R32c)(1)

11. Describe the lights displayed by a vessel underway if engaged on pilotage duty?
(R29a(i)(ii))(2)

12. Describe the additional lights and shape(s) that may be displayed by a power driven vessel if constrained by her draught?(R28)(2)

13. What should a stand-on power-driven vessel which takes action to avoid collision in a crossing situation not do, if the circumstances of the case admit?(**R17c**)(1)

14. A vessel in doubt as to whether she is overtaking or not shall take what action?(**R13c**)(1)

15. Define the term “head-on situation”? (**R14b**)(1)

16. How should a radar equipment be used if fitted and operational? (**R7b**)(1)

17. Describe an isolated danger mark including light characteristics, colour, topmark and the recommended navigable side. (**NP735**)(2)

18. If the circumstances of the case admit, how should any alteration of course and/or speed to avoid collision be? (**R8b**)(1)

Part 2

You are OOW1 in HMS MONTROSE (MTRO) (Length 133m) preparing to sail to start the first Sea Week of BOST; a number of Monday morning Navigation serials are planned and SWO(N) is embarked. Your first serial is Hamoaze Out, 15 minutes before sunrise.

19. What lights should you display when underway? **(R23a)(2)**

20. You also turn on your masthead obstruction lights in accordance with local regulations. State the Rule that justifies this action. **(R1c)(1)**

21. You can safely navigate only within the Hamoaze channel. List 4 types of vessels required not to impede your safe passage? **(R9b, c & d)(2)**

22. At Red 15 range 5c, you detect the Torpoint ferry tracking left to right with a CPA of 1c on the ship's head. In addition, your lookout reports a blue flashing light which leads you to report to the CO that she has right of way due to local regulations by QHM. What does the Rule say about such rules, including their conformity to the IRPC Rules? **(R1b)(1)**

23. After disembarking the pilot in the sound, the SWO(N) informs you that visibility is poor south of the breakwater. When do the Rules regarding restricted visibility apply? **(R19a)(1)**

24. Your SOA for the Rame Head Channel Outbound PILOTEX is 12kts which you are achieving with maximum power on the 3 running Diesel Generators. Will it be appropriate for the Engineers to stand down the fourth engine for routine maintenance noting the prevailing poor visibility? Justify your decision **(R19b)(1)**

25. Give six factors, relating to your operational radar, you will take into account to determine your safe speed. **(R6b)(3)**

26. What actions is required by the Rules if you detect by radar alone the presence of another vessel, include actions to be avoided? **(R19d)(3)**

27. Your lookout reports a sound signal consisting of 1 prolonged blast followed by 2 short blasts at intervals of no more than 2 minutes. List 6 possible types of vessels that could make such signals. **(R35c&d)(3)**

28. Describe your actions if this signal was heard apparently forward of the beam for the first time and is not held on radar. **(R19e)(1)**

29. The PILOTEX ends and the SWO(N) lifts the restriction on visibility? Your next serial is a RAS with RFA WAVE KNIGHT. During your transit to the RV, you sight a vessel at Red 45 at a range of 5nm with a steady CPA. You are 45 degrees on her starboard bow; who is the give way vessel? **(R15)(1)**

30. Describe the initial action of the stand-on vessel in the situation above, and any subsequent action she **may** take if the give-way vessel is not taking appropriate action in accordance with the Rules **(R17a(i)(ii))(2)**

31. Describe the shapes will you display when connected to the tanker. **(R27b(ii))(1)**

32. What lights would you display in addition to that of a power driven vessel if you were conducting the replenishment at night? **(R27b(i))(1)**

33. HMS ARGYLL (ARGL) has a CPA of 2c on the port bow; she is in the WPP as conducting a SOOTAX. Who has right of way in accordance with NATO Rules – MTRO or ARGL?**(ATP 1G Art 140(5))(1)**

34. What does the Rule say about the actions of a vessel engaged in fishing underway towards a vessel restricted in her ability to manoeuvre? **(R18c)(1)**

The RAS concludes successfully and it is followed by SOOTAX to disembark the FOST staff.

Part 2 _____/25

TOTAL _____/50

PART 1

1. How should a vessel normally join or leave a TSS? **(R10b(iii))(1)**

R10b. A vessel using a traffic separation scheme shall:

(iii). normally join or leave a traffic lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable.

2. Describe a masthead light (colour, arc of visibility and position on the ship). **(R21a)(1)**

R21a. Masthead light" means a white light placed over the fore and aft centreline of the vessel showing an unbroken light over an arc of the horizon of 225 degrees and so fixed as to show the light from right ahead to 22.5 degrees abaft the beam on either side of the vessel

3. Nothing in these Rules shall exonerate any vessel, owner, master or crew thereof from what? **(R2a)(1)**

R2a. Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case

4. The Rules regarding lookout, safe speed, risk of collision, action to avoid collision, narrow channels and Traffic Separation Schemes apply in what condition of visibility?**(R4)(1)**

R4. Rules in this section apply in any condition of visibility

5. Describe the lights displayed by a partly submerged vessel or object, or a combination of such vessels or objects being towed if less than 100m in length. **(R24g(i)&(ii))(2)**

R24g. An inconspicuous, partly submerged vessel or object, or combination of such vessels or objects being towed, shall exhibit:

(i). if it is less than 25 metres in breadth, one all-round white light at or near the forward end and one at or near the after end except that dracones need not exhibit a light at or near the forward end;

(ii). if it is 25 metres or more in breadth, two additional all-round white lights at or near the extremities of its breadth;

6. Describe the lights, including any additional lights that may be displayed by a sailing vessel underway, if greater than 20m in length.**(R25b&c)(2)**

R25

(b). A sailing vessel underway shall exhibit:

(i). sidelights;

(ii). a sternlight.

(c). A sailing vessel underway may, in addition to the lights prescribed in paragraph (a) of this Rule, exhibit at or near the top of the mast, where they can best be seen, two all-round lights in a vertical line, the upper being red and the lower green, but these lights shall not be exhibited in conjunction with the combined lantern permitted by paragraph (b) of this Rule.

7. What does the term "vessel not under command" mean? **(R3f)(1)**

R3f. (f). The term “vessel not under command” means a vessel which through some exceptional circumstance is unable to manoeuvre as required by these Rules and is therefore unable to keep out of the way of another vessel.

8. In construing and complying with these Rules, due regard shall be had to what?(**R2b)(2)**

R2b. In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger.

9. Describe the sound signals, including time intervals, required for a vessel at anchor in restricted visibility, if greater than 100m. (**R35g)(2)**

R35g. A vessel at anchor shall at intervals of not more than one minute ring the bell rapidly for about 5 seconds. In a vessel of 100 metres or more in length the bell shall be sounded in the forepart of the vessel and immediately after the ringing of the bell the gong shall be sounded rapidly for about 5 seconds in the after part of the vessel. A vessel at anchor may in addition sound three blasts in succession, namely one short, one prolonged and one short blast, to give warning of her position and of the possibility of collision to an approaching vessel

10. What does the term prolonged blast mean?(**R32c)(1)**

R32c. The term “prolonged blast” means a blast of from four to six seconds’ duration.

11. Describe the lights displayed by a vessel underway if engaged on pilotage duty?
(**R29a(i)(ii)(2)**)

R29(a). A vessel engaged on pilotage duty shall exhibit:

(i). at or near the masthead, two all-round lights in a vertical line, the upper being white and the lower red;

(ii). when underway, in addition, sidelights and a sternlight;

12. Describe the additional lights and shape(s) that may be displayed by a power driven vessel if constrained by her draught?(**R28)(2)**

R28. A vessel constrained by her draught may, in addition to the lights prescribed for power-driven vessels in Rule 23, exhibit where they can best be seen three all-round red lights in a vertical line, or a cylinder

13. What should a stand-on power-driven vessel which takes action to avoid collision in a crossing situation not do, if the circumstances of the case admit?(**R17c)(1)**

R17c. A power-driven vessel which takes action in a crossing situation in accordance with subparagraph (a)(ii) of this Rule to avoid collision with another power-driven vessel shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side.

14. A vessel in doubt as to whether she is overtaking or not shall take what action?(**R13c)(1)**

R13c. When a vessel is in any doubt as to whether she is overtaking another, she shall assume that this is the case and act accordingly

15. Define the term “head-on situation”? **(R14b)(1)**

R14b. Such a situation shall be deemed to exist when a vessel sees the other ahead or nearly ahead and by night she could see the masthead lights of the other in a line or nearly in a line and/or both sidelights and by day she observes the corresponding aspect of the other vessel

16. How should a radar equipment be used if fitted and operational? **(R7b)(1)**

R7b. Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects

17. Describe an isolated danger mark including light characteristics, colour, topmark and the recommended navigable side. **(NP735)(2)**

NP735.

White light: Fl(2) 5s (some variations in period is possible)

Colour: Black with one or more Red horizontal bands

Topmark: Two Black spheres one over the other

Safe water all- round the mark but not too close to the mark

18. If the circumstances of the case admit, how should any alteration of course and/or speed to avoid collision be? **(R8b)(1)**

R8b. Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided

Part 2

You are OOW1 in HMS MONTROSE (MTRO) (Length 133m) preparing to sail to start the first Sea Week of BOST; a number of Monday morning Navigation serials are planned and SWO(N) is embarked. Your first serial is Hamoaze Out, 15 minutes before sunrise.

19. What lights should you display when underway? **(R23a)(2)**

R23a. A power-driven vessel underway shall exhibit:

- (i). a masthead light forward;*
- (ii). a second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 metres in length shall not be obliged to exhibit such light but may do so;*
- (iii). sidelights;*
- (iv). a sternlight.*

20. You also turn on your masthead obstruction lights in accordance with local regulations. State the Rule that justifies this action. **(R1c)(1)**

R1c. Nothing in these Rules shall interfere with the operation of any special rules made by the Government of any State with respect to additional station or signal lights, shapes or whistle signals for ships of war and vessels proceeding under convoy, or with respect to additional station or signal lights or shapes for fishing vessels engaged in fishing as a fleet. These additional station or signal lights, shapes or whistle signals shall, so far as possible, be such that they cannot be mistaken for any light, shape or signal authorized elsewhere under these Rules

21. You can safely navigate only within the Hamoaze channel. List 4 types of vessels required not to impede your safe passage? **(R9b, c & d)(2)**

R9(b). A vessel of less than 20 metres in length or a sailing vessel shall not impede the passage of a vessel which can safely navigate only within a narrow channel or fairway.

(c). A vessel engaged in fishing shall not impede the passage of any other vessel navigating within a narrow channel or fairway.

(d). A vessel shall not cross a narrow channel or fairway if such crossing impedes the passage of a vessel which can safely navigate only within such channel or fairway. The latter vessel may use the sound signal prescribed in Rule 34(d) if in doubt as to the intention of the crossing vessel.

22. At Red 15 range 5c, you detect the Torpoint ferry tracking left to right with a CPA of 1c on the ship's head. In addition, your lookout reports a blue flashing light which leads you to report to the CO that she has right of way due to local regulations by QHM. What does the Rule say about such rules, including their conformity to the IRLC Rules? **(R1b)(1)**

R1b. Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes or inland waterways connected with the high seas and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules.

23. After disembarking the pilot in the sound, the SWO(N) informs you that visibility is poor south of the breakwater. When do the Rules regarding restricted visibility apply? **(R19a)(1)**

R19a. This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility

24. Your SOA for the Rame Head Channel Outbound PILOTEX is 12kts which you are achieving with maximum power on the 3 running Diesel Generators. Will it be appropriate for the Engineers to stand down the fourth engine for routine maintenance noting the prevailing poor visibility? Justify your decision **(R19b)(1)**

No.

R19b. Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility. A power-driven vessel shall have her engines ready for immediate manoeuvre

25. Give six factors, relating to your operational radar, you will take into account to determine your safe speed. **(R6b)(3)**

R6b. Additionally, by vessels with operational radar:

- (i). the characteristics, efficiency and limitations of the radar equipment;*
- (ii). any constraints imposed by the radar range scale in use;*
- (iii). the effect on radar detection of the sea state, weather and other sources of interference;*
- (iv). the possibility that small vessels, ice and other floating objects may not be detected by radar at an adequate range;*
- (v). the number, location and movement of vessels detected by radar;*
- (vi). the more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.*

26. What actions is required by the Rules if you detect by radar alone the presence of another vessel, include actions to be avoided? **(R19d)(3)**

R19d. A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

- (i). an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;*
- (ii). an alteration of course towards a vessel abeam or abaft the beam*

27. Your lookout reports a sound signal consisting of 1 prolonged blast followed by 2 short blasts at intervals of no more than 2 minutes. List 6 possible types of vessels that could make such signals. **(R35c&d)(3)**

R35. In or near an area of restricted visibility, whether by day or night, the signals prescribed in this Rule shall be used as follows:

(c). A vessel not under command, a vessel restricted in her ability to manoeuvre, a vessel constrained by her draught, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in paragraphs (a) or (b) of this Rule, sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts.

(d). A vessel engaged in fishing, when at anchor, and a vessel restricted in her ability to manoeuvre when carrying out her work at anchor, shall instead of the signals prescribed in paragraph (g) of this Rule sound the signal prescribed in paragraph (c) of this Rule.

28. Describe your actions if this signal was heard apparently forward of the beam for the first time and is not held on radar. **(R19e)(1)**

R19e. Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over

29. The PILOTEX ends and the SWO(N) lifts the restriction on visibility? Your next serial is a RAS with RFA WAVE KNIGHT. During your transit to the RV, you sight a vessel at Red 45 at a range of 5nm with a steady CPA. You are 45 degrees on her starboard bow; who is the give way vessel? **(R15)(1)**

The other vessel.

R15. When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel

30. Describe the initial action of the stand-on vessel in the situation above, and any subsequent action she **may** take if the give-way vessel is not taking appropriate action in accordance with the Rules **(R17a(i)(ii))(2)**

R17a.

(i). Where one of two vessels is to keep out of the way the other shall keep her course and speed.
(ii). The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

31. Describe the shapes will you display when connected to the tanker. **(R27b(ii))(1)**

R27b. A vessel restricted in her ability to manoeuvre, except a vessel engaged in mine clearance operations, shall exhibit:

(ii). three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond;

32. What lights would you display in addition to that of a power driven vessel if you were conducting the replenishment at night? **(R27b(i))(1)**

R27b. A vessel restricted in her ability to manoeuvre, except a vessel engaged in mine clearance operations, shall exhibit:

(i). three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white

33. HMS ARGYLL (ARGL) has a CPA of 2c on the port bow; she is in the WPP as conducting a SOOTAX. Who has right of way in accordance with NATO Rules – MTRO or ARGL?**(ATP 1G Art 140(5))(1)**

MTRO (RAS Group) has right of way.

ATP 1G Art 140(5). Ships engaged in launching or recovery of aircraft have the right of way except over ships or formations showing the appropriate signals and are engaged in replenishment (other than VERTREP) or mine countermeasures operations; other ships are to keep clear.

34. What does the Rule say about the actions of a vessel engaged in fishing underway towards a vessel restricted in her ability to manoeuvre? **(R18c)(1)**

R18c. (c). A vessel engaged in fishing when underway shall, so far as possible, keep out of the way of:

(i). a vessel not under command;

(ii). a vessel restricted in her ability to manoeuvre

The RAS concludes successfully and it is followed by SOOTAX to disembark the FOST staff.

Part 2 _____/25

TOTAL _____/50